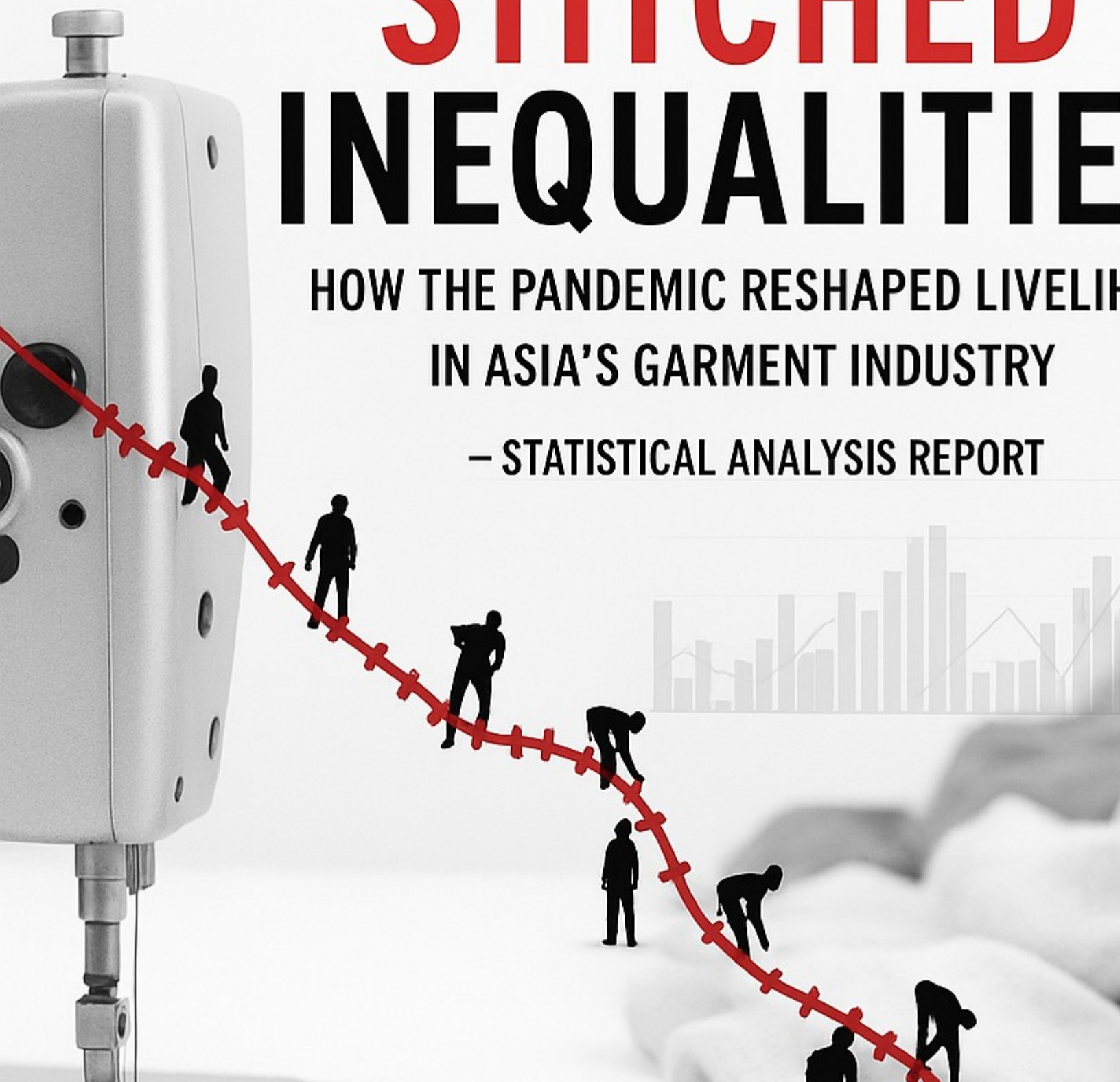


STITCHED INEQUALITY

HOW THE PANDEMIC RESHAPED LIVELIHOODS
IN ASIA'S GARMENT INDUSTRY
— STATISTICAL ANALYSIS REPORT



Impact of COVID-19 on Employment and Livelihood of People

Created & Submitted By: Ravi Pratap Singh

Enrollment Number: M2024BSASS022

Designs and layout: Ravi Pratap Singh

Graphs and tables sourced from SPSS

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Introduction

When the world went into lockdown during the COVID-19 pandemic, headlines spoke about falling GDPs, shrinking exports, and closed borders. But behind those big numbers were millions of quiet stories — workers struggling to put food on the table, families tightening budgets, and industries fighting to survive. One of the industries hit hardest was garments — especially in Asia, where countries like Bangladesh, Cambodia, India, Indonesia, Pakistan, and Sri Lanka rely heavily on clothing exports for jobs and income.

Factories that once buzzed with activity suddenly stood silent. Orders from overseas buyers dried up almost overnight. Ships full of goods were delayed, and workers faced job cuts or reduced hours. As a statistics student, I wanted to put numbers to these struggles — not to replace the human story, but to measure its scale and understand what was really happening.

In this report, I work with data on average wage loss, average consumption loss, and a measure called the Livelihood Stress Index (CLSI) for these six countries. Wage loss tells us how much workers' incomes fell. Consumption loss shows how much their spending dropped — often because they simply couldn't afford essentials. The CLSI combines different signs of stress into one score.

The analysis doesn't just stop at averages. By looking at standard deviations, I could see how varied the impact was between countries. More importantly, I found

something striking — consumption loss and CLSI are almost perfectly linked (correlation of 0.974). That means whenever spending dropped, stress levels rose in nearly the same proportion.

As we move forward, we'll see how these numbers paint a picture of an industry under immense strain. This is not just a story about garments — it's about why understanding data matters when lives are on the line.

Distribution of Respondents by Country: Who Was Counted and Where?

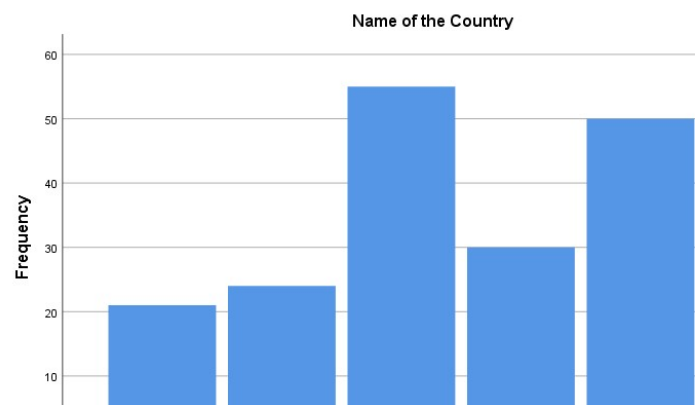


Chart 1: Frequency of Respondents by Country

The first bar chart displays the **frequency of survey respondents by country**. India and Pakistan dominate the respondent base, with 55 and 50 factories, respectively, followed by Indonesia, Cambodia, Bangladesh, and Sri Lanka. This distribution highlights India and Pakistan as key case studies for detailed statistical interpretation while reminding us

to temper national inferences for counties with smaller samples like Sri Lanka.

Key Insight: Countries most present in the data—India, Pakistan, Indonesia—are not only major garment exporters but also among the most impacted by labor disruptions during the pandemic.

Wage and Consumption Losses

Name of the Country		WageLoss	ConsumptionLoss
Bangladesh	Mean	4356.8185	318.6891
	N	21	21
	Std. Deviation	2820.58307	707.96927
Cambodia	Mean	2378.0000	-371337.3755
	N	24	24
	Std. Deviation	5194.74304	133963.54484
India	Mean	3477.4486	19158.6141
	N	55	55
	Std. Deviation	2209.35531	8970.45265
Indonesia	Mean	7749.8897	4676.2836
	N	30	30
	Std. Deviation	11320.28789	16951.90974
Pakistan	Mean	4473.9767	8721.6760
	N	50	50
	Std. Deviation	3791.55734	17476.66929
Sri Lanka	Mean	6264.8459	1404.5419
	N	11	11
	Std. Deviation	5747.96950	5185.38554
Total	Mean	4528.4486	-38009.7303
	N	191	191
	Std. Deviation	5763.54090	135701.06564

Table 1: "Name of the Country | WageLoss & ConsumptionLoss (Mean, Std. Dev, N)"

Wage losses form the cornerstone measure of pandemic impact. According to the wage and consumption loss dataset, Indonesia recorded the most severe average wage decline, at over 7,700 units, followed by Sri Lanka at approximately 6,265. Bangladesh and Pakistan, despite registering wage losses in the 4,300–4,500 range, show relatively narrower variability, suggesting a more uniform level of distress across their respondent samples.

The most striking outlier is Cambodia's **mean consumption loss of –371,337**, a magnitude far beyond any other country, suggesting deep, structural disruptions to household spending capacity. Given the high standard deviation for Cambodia's consumption loss (over 133,000), we can infer that while some workers may have experienced moderate declines, a significant portion faced catastrophic consumption cuts — potentially due to complete income cessation, migration reversals, or debt-driven financial collapse.

From a regional perspective, the aggregate average wage loss stands at ~4,528 units, but high inter-country standard deviations point to **asymmetric exposure to shocks**, reflecting differences in export orientation, government relief measures, and labor-management relations.

Severity of Wage Losses: Mean Percentage Loss by Country

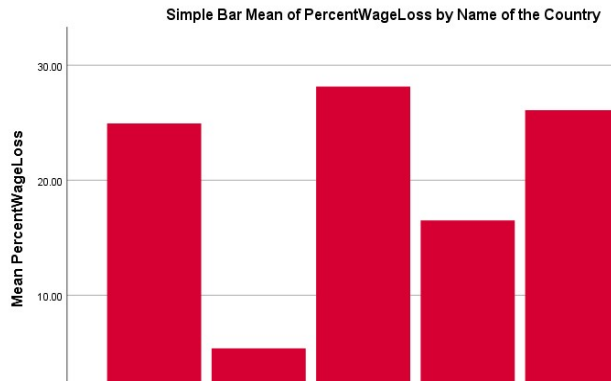


Chart 2: Mean Wage Loss Percentage by Country

The second chart illustrates the **mean percent wage loss** by country, a direct indicator of worker-level financial struggle. India, Pakistan and Bangladesh suffered the highest average wage losses (28–26%), consistent with reports of widespread factory shutdowns and pay suspensions. Cambodia appears as an outlier with notably lower mean wage loss—aligning with some workers receiving government subsidies, though this does not necessarily reflect well-being due to other shocks visible in earlier table-based findings.

Interpretation: While Cambodia’s wage loss seems “better” in this figure, other structural data (such as consumption loss) hint at deeper, hidden vulnerabilities. The starkest wage hits in India, Pakistan, and Bangladesh suggest that payroll-based relief was either less available or less effective in these countries—connecting visual data to actionable policy need.

Wage Inequality and Outliers: Boxplot of

Average Monthly Wage (2020) by Country

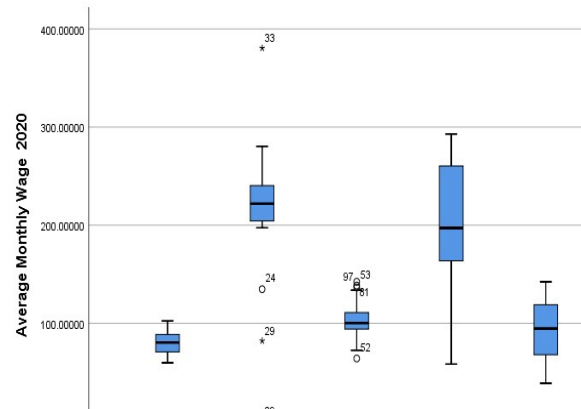


Chart 3: Boxplot of Average Monthly Wage by Country

This boxplot sharply details **wage inequality within and between countries**. Indonesia’s wage distribution is the widest—indicating substantial internal disparity, with several factories or regions far above or below average. India, Cambodia, and Sri Lanka show distinct clusters, with visible outliers above and below the inter-quartile range. Bangladesh and Pakistan cluster toward the lower end of the wage scale, reflective of their status as low-wage garment hubs.

Critical Analysis: The appearance of multiple extreme outliers shows that even within one country or sector, pandemic impacts were not uniform. Individual factories or workers fell far outside typical wage bounds, perhaps due to localized lockdowns, bankruptcy, or particular patterns of supply chain disruption. This visual evidence underlines the broader finding of the earlier tables: averages alone mask critical pockets of extreme hardship.

Wage Recovery Patterns

Name of the Country * Category of Wage Recovery Crosstabulation

		Category of Wage Recovery			Total
		1.00	2.00	3.00	
Name of the Country	Bangladesh	0	20	1	21
	Cambodia	1	2	20	23
	India	2	45	8	55
	Indonesia	2	13	15	30
	Pakistan	1	28	21	50
	Sri Lanka	1	7	3	11
Total		7	115	68	190

Table 2: "Name of the Country * Category of Wage Recovery Crosstabulation"

To move beyond losses, the wage recovery dataset categorizes workers into three recovery outcomes: ($\leq .499$: **No Recovery**, $.50-.899$ **Partial Recovery**, and $\geq .9$ **Full Recovery**.)

Partial recovery is the modal outcome, encompassing 60% of respondents across the sample. This suggests that while some form of income stream was restored for the majority of workers, it fell short of pre-pandemic levels — with implications for prolonged vulnerability. Cambodia's exceptional case, where over 85% of workers reported **full recovery**, stands in contrast to Bangladesh and Sri Lanka, where full recovery barely crosses single-digit instances.

The interpretation here is nuanced: high rates of full recovery do not automatically equate to overall wellbeing. For instance, if full wage recovery occurred alongside increased household debt or inflationary pressures, the net livelihood security could remain low.

Correlational Dynamics Between Crisis Indicators

		Correlations				
		PercentWageLoss	PercentEmploymentLoss	PercentConsumptionLoss	WageRecoveryRatio	Covid Livelihood Stress Index
PercentWageLoss	Pearson Correlation	1	.846**	.336**	-.826**	.407**
	Sig. (2-tailed)		0.000	0.000	0.000	0.000
	N	190	190	190	190	190
PercentEmploymentLoss	Pearson Correlation	.846**	1	.425**	-.747**	.493**
	Sig. (2-tailed)	0.000		0.000	0.000	0.000
	N	190	190	190	190	190
PercentConsumptionLoss	Pearson Correlation	.336**	.425**	1	-.301**	.974**
	Sig. (2-tailed)	0.000	0.000		0.000	0.000
	N	190	190	191	190	191
WageRecoveryRatio	Pearson Correlation	-.826**	-.747**	-.301**	1	-.361**
	Sig. (2-tailed)	0.000	0.000	0.000		0.000
	N	190	190	190	190	190
Covid Livelihood Stress Index	Pearson Correlation	.407**	.493**	.974**	-.361**	1
	Sig. (2-tailed)	0.000	0.000	0.000	0.000	
	N	190	190	191	190	191

** . Correlation is significant at the 0.01 level (2-tailed).

Table 3: "Correlations – PercentWageLoss, PercentEmploymentLoss, PercentConsumptionLoss, WageRecoveryRatio, Covid Livelihood Stress Index"

The correlation matrix provides a systems-level view of how shocks interacted:

- **Wage loss and employment loss** are tightly linked ($r = 0.846$, $p < 0.01$), confirming that when jobs disappeared, wages also suffered proportionally.
- **Consumption loss and the Livelihood Stress Index (CLSI)** share a near-perfect positive correlation ($r = 0.974$), establishing reduced spending as the single

strongest predictor of self-reported stress.

- The **Wage Recovery Ratio** strongly and negatively correlates with wage loss ($r = -0.826$), reaffirming that those who suffered greater losses were less able to recover wages — a cyclical entrapment.

These values statistically confirm that the pandemic's impact was **multidimensional** and that interventions must target the chain reaction, not isolated variables.

Resilience and Recovery: The Wage-Consumption Recovery Relationship

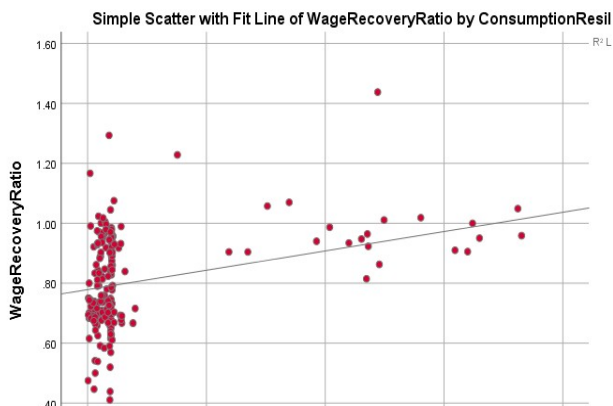


Chart 4: Scatterplot of Wage Recovery Ratio vs. Consumption Resilience Index

Perhaps the most sophisticated chart, this scatterplot examines the **relationship between wage recovery ratios and the Consumption Resilience Index**—a composite measure of how well households sustained spending power during the crisis. The slight positive trendline ($R^2 = 0.091$)

indicates a weak but positive connection: in places or instances where household consumption was more resilient, wage recovery was slightly higher.

Interpretive Outcome: The high dispersion and low R^2 summarize a crucial pandemic insight: **recovery was highly individualistic**. Factors such as household savings, remittance flows, food support, and local economic conditions all shaped how recovery materialized. The absence of a strong correlation means policy must address both wage and consumption dimensions separately, not assuming that improvement in one reliably guarantees the other.

The Web of Shocks: Scatterplot Matrix of Income, Employment, Consumption & Stress

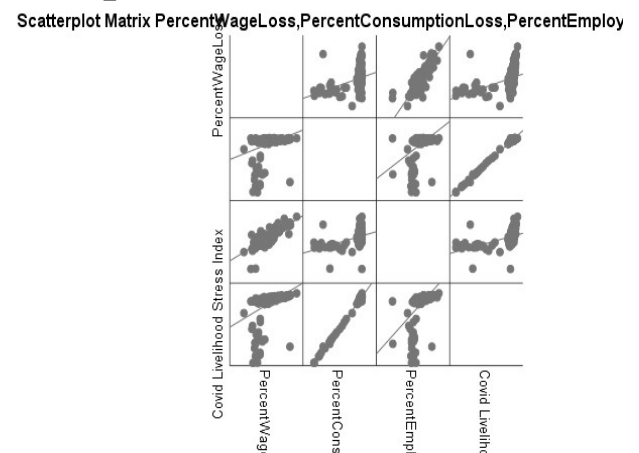


Chart 5: Scatterplot Matrix of Percent Wage Loss, Consumption Loss, Employment Loss, Stress Index

This scatterplot matrix displays the relationships among the core pandemic variables for each respondent, revealing strong positive alignments: those who lost more wages typically reported both higher employment loss and greater overall stress. The clearest trend is between consumption loss and the stress index—confirming that loss of spending power drives acute distress.

Yet, the matrix also shows some scatter between income and consumption variables, hinting at the role of buffers: savings, government aid, or community support softened income shocks for some households but not all. Virtually no households escape the crisis untouched—nearly all data points fall far from a “no loss, no stress” origin.

Mapping the Depth of Distress: Distribution of the Covid Livelihood Stress Index

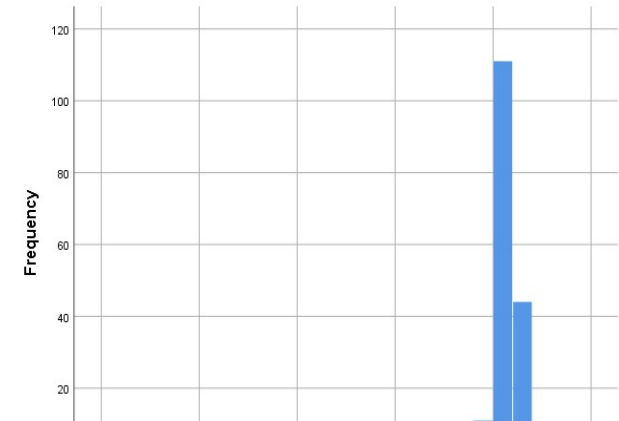


Chart 6 : Histogram of Covid Livelihood Stress Index

This histogram reveals how the pandemic’s structural shocks translated into lived stress for garment workers. The overwhelming majority of respondents cluster around a stress index near zero, creating a tall, narrow central peak—signaling that most experienced moderate but substantial hardship. However, the chart’s most telling feature is its long negative tail, with some households experiencing extreme stress indices below –800. This minority, though statistically less numerous, signals grave and persistent vulnerability—likely the result of total job/income loss, debt burdens, or minimal access to relief.

The reported average (mean = –24.71) and exceptionally high standard deviation (~143) reflect both overall hardship and immense diversity in impact. The distribution’s skew confirms that while “average” hardship is instructive, targeted policy must address the blighted minority who suffered the most.

Severity of Employment Shock and Livelihood stress

Report			
Name of the Country		Employment Shock Severity- mean of standardized Z Score	Covid Livelihood Stress Index
Bangladesh	Mean	0.1552	16.4701
	N	21	21
	Std. Deviation	0.39451	7.88552
Cambodia	Mean	-1.5516	-376.6625
	N	24	24
	Std. Deviation	0.56141	134.63845
India	Mean	0.3152	36.8155
	N	55	55
	Std. Deviation	0.38226	12.98590
Indonesia	Mean	0.0443	15.2014
	N	30	30
	Std. Deviation	0.50829	11.34021
Pakistan	Mean	0.2778	26.5076
	N	50	50
	Std. Deviation	0.82826	26.43810
Sri Lanka	Mean	0.0509	15.3219
	N	11	11
	Std. Deviation	0.70712	12.65149
Total	Mean	-0.0045	-24.7079
	N	191	191
	Std. Deviation	0.83366	142.90487

Table 4: "Employment Shock Severity (Z Score) | Covid Livelihood Stress Index"

Employment shock severity, computed using standardized Z-scores, indicates that Bangladesh, India, and Pakistan experience above-average employment shocks relative to the region. Cambodia registers an unusually negative Z-score (−1.55), which, combined with its dismal consumption loss profile, paints a picture where **income and livelihood collapse occurred via different pathways** — not necessarily through mass layoffs, but possibly through underemployment, wage suspension, or contract non-renewals.

When paralleled with the CLSI values, India stands out with the **highest positive stress score (+36.8)**, suggesting both job and consumption shocks translated more intensively into perceived livelihood strain compared to other countries.

Poverty Impact Disparities

Multiple Comparisons					
Dependent Variable: PovertyImpactFactor					
Tukey HSD					
(I) Code for countries		Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval Lower Bound Upper Bound
Bangladesh	Cambodia	.58719 [*]	0.05144	0.000	0.4390 0.7353
	Indonesia	.49292 [*]	0.04898	0.000	0.3519 0.6340
	India	0.07173	0.04416	0.584	-0.0554 0.1989
	Pakistan	0.07170	0.04477	0.598	-0.0572 0.2006
	Sri Lanka	0.14714	0.06408	0.201	-0.0374 0.3317
Cambodia	Bangladesh	-.58719 [*]	0.05144	0.000	-0.7353 -0.4390
	Indonesia	-0.09428	0.04715	0.347	-0.2301 0.0415
	India	-.51546 [*]	0.04212	0.000	-0.6368 -0.3942
	Pakistan	-.51549 [*]	0.04275	0.000	-0.6386 -0.3924
	Sri Lanka	-.44006 [*]	0.06268	0.000	-0.6206 -0.2595
Indonesia	Bangladesh	-.49292 [*]	0.04898	0.000	-0.6340 -0.3519
	Cambodia	0.09428	0.04715	0.347	-0.0415 0.2301
	India	-.42119 [*]	0.03907	0.000	-0.5337 -0.3087
	Pakistan	-.42121 [*]	0.03976	0.000	-0.5357 -0.3067
	Sri Lanka	-.34578 [*]	0.06068	0.000	-0.5205 -0.1710
India	Bangladesh	-0.07173	0.04416	0.584	-0.1989 0.0554
	Cambodia	.51546 [*]	0.04212	0.000	0.3942 0.6368
	Indonesia	.42119 [*]	0.03907	0.000	0.3087 0.5337
	Pakistan	-0.00002	0.03364	1.000	-0.0969 0.0969
	Sri Lanka	0.07541	0.05686	0.770	-0.0883 0.2392
Pakistan	Bangladesh	-0.07170	0.04477	0.598	-0.2006 0.0572
	Cambodia	.51549 [*]	0.04275	0.000	0.3924 0.6386
	Indonesia	.42121 [*]	0.03976	0.000	0.3067 0.5357
	India	0.00002	0.03364	1.000	-0.0969 0.0969
	Sri Lanka	0.07543	0.05733	0.776	-0.0897 0.2405
Sri Lanka	Bangladesh	-0.14714	0.06408	0.201	-0.3317 0.0374
	Cambodia	.44006 [*]	0.06268	0.000	0.2595 0.6206
	Indonesia	.34578 [*]	0.06068	0.000	0.1710 0.5205
	India	-0.07541	0.05686	0.770	-0.2392 0.0883
	Pakistan	-0.07543	0.05733	0.776	-0.2405 0.0897

*. The mean difference is significant at the 0.05 level.

Table 5a: "Multiple Comparisons (Tukey HSD) for Poverty Impact Factor"

PovertyImpactFactor

Tukey HSD_{a,b}

Code for countries	N	Subset for alpha = 0.05		
		1	2	3
Cambodia	24	0.0601		
Indonesia	30	0.1544		
Sri Lanka	11		0.5002	
India	55		0.5756	0.5756
Pakistan	50		0.5756	0.5756
Bangladesh	21			0.6473
Sig.		0.411	0.657	0.704

Means for groups in homogeneous subsets are displayed.

a. Uses Harmonic Mean Sample Size = 23.837.

b. The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not guaranteed.

Table 5b: "PovertyImpactFactor – Homogeneous Subsets"

Tukey's HSD test of mean differences in the Poverty Impact Factor identifies three statistically meaningful clusters:

- **Lowest income security:** Cambodia (mean 0.0601) and Indonesia (0.1544) — significant differences from Bangladesh, India, and Pakistan ($p < 0.001$).
- **Middle tier:** Sri Lanka, India, and Pakistan hover around 0.50–0.57, indicating moderate deterioration.
- **Best-performer subset:** Bangladesh (0.6473) statistically outperforms other nations in poverty impact, though the absolute figure still indicates substantive adverse effects.

These results demonstrate **stratified vulnerability** — Cambodia's poverty escalation is distinct both in magnitude and statistical significance.

Factory-Level Extreme Losses

Case Summaries^a

	Factory ID	Name of the Country	WageLoss
1	IDN001	Indonesia	56427.59
2	IDN059	Indonesia	23980.07
3	IDN004	Indonesia	23903.10
4	CAM009	Cambodia	23697.00
5	IDN006	Indonesia	21760.07
6	SL004	Sri Lanka	17300.00
7	SL007	Sri Lanka	14513.68
8	PK107	Pakistan	13934.55
9	PK102-10	Pakistan	13719.92
10	PK055	Pakistan	12571.80
Total	N	10	10

a. Limited to first 10 cases.

Table 6: "Case Summaries – Top 10 Factories by WageLoss"

Micro-level analysis reveals **geographically concentrated hotspots of wage collapse**. Indonesia dominates the top 10 loss cases, with one facility (IDN001) reporting wage losses of 56,427 — over **12 times** the regional average and far beyond standard deviations observed in national aggregates.

Such outliers are critical for policy and buyer interventions because they likely result from single large contracts being cancelled or prolonged export market inactivity. Cambodia appears once (CAM009) on the list, confirming the general narrative of selective but massive disruptions.

Interaction of Wage Recovery and Stress

Report			
Category of Wage Recovery		PercentWageLoss	Covid Livelihood Stress Index
1.00	Mean	61.3087	-10.9858
	N	7	7
	Std. Deviation	7.41016	181.05601
2.00	Mean	29.1103	22.2221
	N	115	115
	Std. Deviation	12.91155	54.64290
3.00	Mean	6.6832	-96.6604
	N	68	68
	Std. Deviation	10.86513	188.32061
Total	Mean	22.2700	-21.5487
	N	190	190
	Std. Deviation	17.78964	136.43136

Table 7: "Category of Wage Recovery | PercentWageLoss Mean | Covid Livelihood Stress Index Mean"

This dataset allows examination of how recovery categories interact with both wage loss magnitudes and stress levels:

- **No Recovery group:** Mean wage loss at 61.31% indicates devastating income shocks, yet their stress index (-10.99) is not the most negative — potentially reflecting adaptation strategies or social networks cushioning immediate stress.
- **Partial Recovery group:** Wage loss reduced to 29%, with a positive CLSI score (+22.22) suggesting ongoing strain despite some recovery.

- **Full Recovery group:** Lowest wage loss (~6.68%) but paradoxically worst average stress score (-96.66), implying that factors beyond income restoration (such as debt, inflation, or health expenses) are driving livelihood insecurity.

This paradox underscores the **multifactorial nature of recovery** — wages alone do not ensure resilience.

Conclusion

After analysing the data, one reality became clear: the pandemic didn't just slow down production — it shook the very foundations of livelihoods. Countries with large garment sectors like India, Pakistan, and Indonesia faced some of the steepest wage and consumption losses, proving how vulnerable export-dependent economies can be when crises strike.

The near-perfect correlation between consumption loss and livelihood stress is more than a statistical fact — it's a warning. It shows that tracking household spending could help governments and organisations identify distress early and respond faster in future crises.

For me as a student, this project reinforced that statistics aren't just about complex formulas — they're a way to uncover patterns that have very real human consequences. In a crisis, these numbers can give us more than facts; they can give us direction, helping turn damage into lessons and lessons into action.

Declaration of AI

Different AI have been used in making this report.

1. Chatgpt is used for rearranging the variables and providing the idea that which variables from the spss can be best used in the topic “Stitched Inequalities: How the Pandemic Reshaped Livelihoods in Asia’s Garment Industry – Statistical Analysis Report”.
2. Chatgpt is also used sometime when I was not getting the correct output so chatgpt was telling me what mistakes I made sometime putting the variable list in dependent instead of independent list etc.
3. Perplexity is used for writing the introduction and conclusion in word limit.
4. Perplexity is also used for writing the report in a presentable manner like what to write in pointers and paragraph.

[The End]